

X_1 (Working hour)	X_2 (Sleep hours)	Y (Performance)
1	6	60
2	7	65
3	8	70

X_1 (Performance)	X_2 Working hours	X_3 Sleep hours
60	1	6
65	2	7
70	3	8

X_1	X_2	X_3	$X_1 X_2$	$X_1 X_3$	$X_2 X_3$	$(X_2)^2$	$(X_3)^2$
60	1	6	60	360	6	1	36
65	2	7	130	455	14	4	49
70	3	8	210	560	24	9	64
$\sum X_1 = 195$	$\sum X_2 = 6$	$\sum X_3 = 21$	$\sum X_1 X_2 = 400$	$\sum X_1 X_3 = 1375$	$\sum X_2 X_3 = 44$	$\sum (X_2)^2 = 14$	$\sum (X_3)^2 = 149$

$$\begin{aligned}\sum X_1 &= N a_{1.23} + b_{12.3} \sum X_2 + b_{13.2} \sum X_3 \\ \sum X_1 X_2 &= a_{1.23} \sum X_2 + b_{12.3} \sum (X_2)^2 + b_{13.2} \sum X_2 X_3 \\ \sum X_1 X_3 &= a_{1.23} \sum X_3 + b_{12.3} \sum X_2 X_3 + b_{13.2} \sum (X_3)^2\end{aligned}$$

$$\begin{aligned}195 &= 3 a_{1.23} + 6 b_{12.3} + 21 b_{13.2} \\ 400 &= 6 a_{1.23} + 14 b_{12.3} + 44 b_{13.2} \\ 1375 &= 21 a_{1.23} + 44 b_{12.3} + 149 b_{13.2}\end{aligned}$$

(Infinite Solutions) Independent variables are perfectly correlated. Perfect multicollinearity therefore

One of the possible solution is
Multiply eq 1 by 2

$$\begin{aligned}390 &= 6 a_{1.23} + 12 b_{12.3} + 42 b_{13.2} \\ \text{Subtract from eq 2} \\ 400 - 390 &= 6 a_{1.23} - 6 a_{1.23} + 14 b_{12.3} - 12 b_{12.3} + 44 b_{13.2} - 42 b_{13.2}\end{aligned}$$

$$10 = 2b_{12 \cdot 3} + 2b_{13 \cdot 2}$$

$$10 = 2(b_{12 \cdot 3} + b_{13 \cdot 2})$$

$$\boxed{b_{12 \cdot 3} + b_{13 \cdot 2} = 5}$$

Multiply eq 1 by 7

$$1365 = 21a_{1 \cdot 23} + 42b_{12 \cdot 3} + 147b_{13 \cdot 2}$$

Subtract from eq 3

$$1375 - 1365 = 21a_{1 \cdot 23} - 21a_{1 \cdot 23} + 44b_{12 \cdot 3} - 42b_{12 \cdot 3} + 149b_{13 \cdot 2} - 147b_{13 \cdot 2}$$

$$10 = 2b_{12 \cdot 3} + 2b_{13 \cdot 2}$$

$$10 = 2(b_{12 \cdot 3} + b_{13 \cdot 2})$$

$$\boxed{b_{12 \cdot 3} + b_{13 \cdot 2} = 5}$$

Both eq gives $b_{12 \cdot 3} + b_{13 \cdot 2} = 5$

Thus coefficient are not unique

As we know

$$b_{12 \cdot 3} + b_{13 \cdot 2} = 5$$

So probable solutions

$b_{12 \cdot 3}$	$b_{13 \cdot 2}$	
1	4	$1+4=5$
2	3	$2+3=5$
3	2	$3+2=5$
4	1	$4+1=5$

So we need to find $a_{1 \cdot 23}$

We use eq 1

$$195 = 3a_{1 \cdot 23} + 6b_{12 \cdot 3} + 21b_{13 \cdot 2}$$

$$b_{12 \cdot 3} = 2 \text{ and } b_{13 \cdot 2} = 3$$

$$195 = 3a_{1 \cdot 23} + 6(2) + 21(3)$$

$$195 = 3a_{1 \cdot 23} + 12 + 63$$

$$195 = 391.23 + 75$$

$$195 - 75 = 391.23$$

$$120 = 391.23$$

$$\frac{120}{3} = 91.23$$

3

$$\boxed{91.23 = 40}$$

Final Regression equation
for this solution is

$$X_1 = 40 + 2X_2 + 3X_3$$

Prediction (4 hours & sleep 9 hours
work)

So

$$X_1 = 40 + 2(4) + 3(9)$$

$$X_1 = 40 + 8 + 27$$

$$\boxed{75}$$